Rithwik Thirumalaraju

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Objective

AI Engineer with hands-on experience in Generative AI, LLM-powered systems, and Retrieval-Augmented Generation (RAG), aiming to build scalable AI services that transform natural language into structured intelligence. Passionate about developing production-ready AI infrastructure that delivers measurable business impact.

Skills

Languages: Python
Core Skills: Machine Learning, Deep Learning, NLP, Generative AI, Agentic AI
Frameworks & Libraries: TensorFlow, OpenCV, LangChain, RAG
Web Technologies: Flask, React(basic)
Frontend Basics: HTML, CSS
Databases: ArangoDB, MongoDB, SQL
Applications & Tools: Streamlit
Containerization & Orchestration: Docker (basic), Kubernetes (basic)
AI Development Tools: Cursor, Kiro, Windsurf, Qoder, Anti-Gravity, Claude Code CLI, Gemini CLI

Experience

AI/ML Engineer, ArangoAI (contract) Jul 1, 2025 – Jul 1, 2026

- Worked on Natural Language Services focused on **Text-to-AQL (txt2aql)**.
- Contributed to **gRPC service** that translates natural language queries into ArangoDB Query Language (AQL), executes queries on ArangoDB, and explains the results back in natural language.
- Integrated OpenAI models for query generation and explanation; designing architecture to support multiple LLM providers including self-hosted LLMs.
- Contributed to **ArangoGraphML** – a RESTful API service that enables users to train, manage, and deploy GraphML models backed by ArangoDB.
- Worked on scalable backend architecture, API design, and LLM-based query validation for production-grade graph intelligence systems.

AI Engineer, BluePal Software Solutions, Hyderabad Aug 2024 – Present

- Designed and developed AI-powered applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic AI workflows.
- Built intelligent chatbots using the Groq API with open-source LLMs, integrating document retrieval, conversational memory, and context-aware responses.
- Developed RAG pipelines using vector databases and embedding models to enable accurate semantic search and knowledge-based question answering.
- Integrated REST APIs and backend services using Python and Flask/FastAPI to support AI-powered applications.
- Built automation workflows by leveraging OpenManus, developing custom wrappers to automate repetitive tasks and orchestrate multi-step AI agent execution.
- Worked with prompt engineering, tool calling, and API integrations to improve response quality and automate business processes.
- Collaborated with cross-functional teams to design, develop, test, and deploy scalable AI-enabled backend services.

Internships

SAK Informatics Jan 2024 – Apr 2024

- Developed CNN-based deep learning model integrated with OpenCV to recognize hand signs and output corresponding word/letter.
- Implemented real-time detection pipeline with live predictions.

Indian Servers Apr 2023 – Jun 2023
- Fine-tuned Hugging Face pre-trained model for accurate movie recommendations.
- Integrated model with Telegram bot API for seamless user interaction.

1stop Aug 2021 – Oct 2021
- Built supervised learning model to detect fake news from large datasets.
- Designed misinformation filtering mechanism for indistinguishable news sources.

Education

B.Tech in Computer Science and Engineering, Sree Venkateswara College of Engineering (JNTUA) 2024

Certifications

- NVIDIA – Building LLM Applications using Prompt Engineering
- ArangoDB – ArangoDB Certified Professional
- NPTEL – Cloud Computing Basics
- Anthropic – Claude Code in Action

Declaration

I hereby state that all facts mentioned above are true to the best of my knowledge.